

STAT5003 Group 07 — A2 Plan

NSW Road Crash Severity Prediction | Workshop 12

Section 1: 20-Slide Presentation Plan

Slide #	Title	Content Source
1	Title Slide	Static: group name, date, subject code
2	Research Question & Motivation	Static text: RQ statement, fatalities stat, stakeholder value
3	Dataset Overview	Table from summary stats :170,962 records, 59 variables, CRASH + TRAFFIC UNIT merged on Crash ID
4	Class Imbalance	ggplot bar chart - Fatal ~1.4%, Injury, Non-casualty class proportions
5	Key EDA(Speed vs Severity)	ggplot grouped bar - fatal proportion by speed band (<=50, 60-80, 90-100, 110+)
6	Key EDA (High-Risk Scenarios)	flextable - top 10 high-risk combinations (speed limit x lighting x urbanisation) from A1
7	Temporal Patterns	ggplot - crash frequency by hour of day and day of week
8	Preprocessing Pipeline	Bullet points - 9-step pipeline overview
9	Feature Engineering	Table - is_weekend, is_rush_hour, is_night, is_wet, speed bands, has_motorcycle, has_heavy_vehicle
10	Handling Class Imbalance (SMOTE)	Before/after bar chart of class distribution; SMOTE applied inside training fold only
11	Model Overview	Summary table - 6 models, justification, R package (nnet, rpart, randomForest, xgboost, kkn, e1071)
12	Logistic Regression Results	3x3 confusion matrix heatmap + Macro-F1, Fatal Recall, Weighted F1, Kappa
13	Decision Tree Results	rpart.plot tree diagram + confusion matrix heatmap
14	Random Forest Results	Variable importance plot (top 15) + confusion matrix heatmap + metrics
15	XGBoost Results	Variable importance plot (by Gain) + training loss curve + confusion matrix
16	KNN + Naive Bayes Results	Side-by-side confusion matrices + metric comparison for both models
17	Model Comparison	flextable — all 6 models x Macro-F1, Fatal Recall, Weighted F1, Cohen's Kappa

Slide #	Title	Content Source
18	Best Model & Answer to RQ	Best model identified; key predictive features; direct answer to research question
19	Conclusion	Bullet points — summary of findings, practical implications for TfNSW / emergency responders
20	Reflection & Future Scope	Limitations (imbalance, no spatial model, cross-sectional data); future directions (deep learning, real-time, spatial modelling)

Section 2: Per-Model Plot Plans

Each model should produce the plots listed below. All confusion matrices must annotate Fatal Recall explicitly.

Multinomial Logistic Regression

Plot	Purpose
Confusion matrix heatmap (3x3)	See misclassification direction across Fatal / Injury / Non-casualty
Coefficient plot (log-odds per class)	Interpretability — which features drive each severity class
Per-class ROC curve (one-vs-rest)	How well the model separates each class

Decision Tree

Plot	Purpose
Tree diagram (rpart.plot)	Full interpretability — decision rules that lead to Fatal prediction
Confusion matrix heatmap	Misclassification pattern
CP (complexity parameter) plot	Shows where pruning was applied during cross-validation

Random Forest

Plot	Purpose
Variable importance plot (top 15)	Core finding — what features predict crash severity most strongly
OOB error vs number of trees	Shows model stabilisation point; confirms ntree is sufficient
Confusion matrix heatmap	Misclassification pattern

XGBoost

Plot	Purpose
Variable importance plot (by Gain)	Compare with RF — do both models agree on top features?
Training loss curve (rounds vs log-loss)	Shows convergence and absence of overfitting
Confusion matrix heatmap	Misclassification pattern

KNN

Plot	Purpose
k vs Macro-F1 curve	Shows optimal k selection from cross-validation; justifies chosen k
Confusion matrix heatmap	Misclassification pattern

Naive Bayes

Plot	Purpose
Confusion matrix heatmap	Misclassification pattern
Predicted probability distributions by true class	Calibration check — Naive Bayes often over-confident on minority classes

Section 3: Comparison Plots (Slides 17-18)

Plot	Purpose
Metric comparison bar chart (6 models x 3 metrics)	Primary model selection visual — Macro-F1, Fatal Recall, Weighted F1 side-by-side
Best model 3x3 confusion matrix with Fatal Recall annotated	Final answer visual — shows absolute classification performance
Variable importance overlay — RF vs XGBoost top 10	Do both models agree on top predictors? Directly answers the RQ

Note: For all confusion matrices, annotate Fatal Recall explicitly — models below 0.50 Fatal Recall are deprioritised regardless of overall F1 score.